

LE HAI TRUNG

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GitHub: [Vaderplayz](#) | [Online Portfolio](#)

About Me

Mechanical Engineering undergraduate and research assistant at VinUniversity. I like to implement a product-based research and development method, whether it is scientific research or solving engineering problems. I am interested in real life projects related to soft robotics, UAV or autonomous systems that combine mechanical design, electronic/embed integration and functional prototype.

Education

VinUniversity, College of Engineering and Computer Science

Sep 2023 – Present

Bachelor of Mechanical Engineering / Hanoi, Vietnam

- Cumulative GPA: 3.51/4.00 (as of September 2026).
- Recipient of a 70% merit-based scholarship with an additional 20% financial support; Dean's List, Spring 2025 and Fall 2026.

Research Interests

Robotics and mechatronics; Soft robotics; UAV autonomy; Autonomous systems.

Research Experience

Undergraduate Research Assistant – Biorobotics Laboratory, VinUniversity

2023 – Present

Supervisor: Dr. Thai Mai Thanh / Undergraduate Research Opportunities Program (UROP)

Double-Attack Mechanical Thrombectomy Device

First Author — Accepted, ICARCV 2026

- Developed a proof-of-concept thrombectomy device inspired by the milli-spinner mechanism, combining rotational clot interaction with rapid axial reciprocation generated by a soft hydraulic muscle.
- Designed a micro-hydraulic turbine mechanism to drive rotation while the soft actuator produced simultaneous push-pull motion.
- Experimentally demonstrated up to approximately 7% improvement in localized suction performance, with greater improvement at lower rotational speeds.

A Soft-Actuator-Integrated Thrombectomy Catheter for High-Speed Distal Rotation and Controlled Localized Axial Elongation

First Author — Submitted, ICRA 2027

Hybrid Soft Muscle-Tendon Glove for Post-Stroke Hand Rehabilitation

Co-First Author — Manuscript under review at Sensors and Actuators A

- Designed the overall soft robotic exoskeleton glove and fabricated the soft-muscle actuators for assisted finger motion.
- Developed actuators capable of more than 100% elongation and over 2.5 N output force.
- Fabricated and tested the complete prototype, demonstrating assistance for lifting objects up to 350 g in daily-living-type tasks.

Teleoperated Humanoid Robotic Hand Using Soft Filament Sensors and Soft Muscles

First Author — Manuscript submitted to Journal of Intelligent & Robotic Systems

- Developed the complete teleoperation system integrating lab-developed soft filament sensors, tendon transmission, and hand-fabricated soft-muscle actuation in a 3D-printed humanoid robotic hand.
- Designed and tested the servo-based control architecture in MATLAB/Simulink and evaluated system response using a high-speed camera.
- Demonstrated responsive hand teleoperation with an end-to-end delay below 200 ms.

Industry Experience

UAV Engineering Intern – Phenikaa-X, Drone Team

Jul 2025 – Oct 2025

Hanoi, Vietnam

- Designed, 3D-printed, and flight-tested a battery-holder for the Holybro X500 UAV, ensuring compatibility with the existing platform.
- Implemented and tested PX4 offboard control for GPS-denied flight using AprilTag-based vision localization; conducted circular and figure-eight velocity-control flights.
- Supported onboard camera deployment and data collection and served as a test pilot during flight experiments.

Selected Engineering Projects

[Personal Project] Integrated UAV Platform for Indoor SLAM, 3D Reconstruction, and AprilTag Precision Landing 2024 – Present

Ongoing

- Developed a dual-2D-LiDAR architecture using one horizontal sensor for indoor SLAM and a vertically mounted sensor for environmental scanning and 3D reconstruction in GPS-denied environments.
- Built a reconstruction pipeline that combines sequential 2D scans with UAV pose estimates to generate a stitched 3D point cloud; validated the approach in PX4 SITL and RViz using lawnmower, spiral, and diagonal survey trajectories.
- Deployed the horizontal-LiDAR SLAM pipeline on a custom 10-inch Holybro F450-based UAV equipped with a Raspberry Pi 5 and 2 RPLIDAR sensors.
- Developed and physically demonstrated an indoor AprilTag precision-landing pipeline using local-frame localization, PX4 offboard control, autonomous approach, touchdown, and disarming without GPS.
- Currently integrating the vertical LiDAR, improving point-cloud quality, and extending the physical platform toward complete 3D reconstruction.

[Course Project] Custom Quadcopter Using Teensy 4.1 and MPU6050 2025 – 2026

Intelligent Physical Systems

- Led the design and construction of an X-frame quadcopter, covering electronics layout, airframe design, integration, and PID-based flight control.
- Built a multi-axis steel-rod-and-bearing test rig and diagnosed vibration-induced IMU errors in the initial 3D-printed frame.
- Redesigned and validated a carbon-fiber airframe using ANSYS, then fabricated, assembled, tuned, and successfully flew the revised platform.

[Course Project] Automatic Tool-Interchange System for a Humanoid Robotic Arm 2025 – 2026

Robotics

- Led the design of a rotary tool station inspired by CNC tool changers, enabling a humanoid robotic arm to autonomously select and exchange task-specific tools.
- Developed an electromagnetic coupling interface with integrated power-contact pins and implemented the alignment sequence required for reliable mechanical and electrical engagement.
- Successfully demonstrated autonomous tool exchange on the physical robotic system.

[Course Project] Autonomous Whiteboard Pen Refill Machine 2025 – 2026

Mechanical Synthesis

- Led project planning and overall system design for an automated machine intended to extend marker life and reduce plastic waste.
- Designed and integrated the electrical and motion-control system using stepper motors, drivers, and controllers.
- Designed the CNC-machined acrylic enclosure and served as one of two primary fabricators and assemblers of the functional prototype.

[Course Project] Hand-Gesture-Controlled Differential Mobile Robot for Environmental Surveying 2024 – 2025

Mechatronics

- Developed a differential-drive robot controlled by hand gestures captured through an IMU-equipped glove.
- Implemented head-motion control of a servo-driven camera gimbal using IMU-equipped glasses for remote visual inspection.
- Integrated ESP32 controllers, NRF24 wireless communication, an onboard LCD, and sensors for temperature, humidity, and hazardous-smoke monitoring.

Additional Projects

Wireless Water-Quality Monitoring and Mapping System (pH, TDS, turbidity, temperature, wireless transmission, and Google Maps API); SCARA Robot Kinematics and 2D MATLAB Trajectory Simulation; Buggy Car Chassis Design and Preliminary ANSYS Structural Analysis; 5-Inch ArduPilot Quadcopter; Mini Hawk VTOL Tilt-Rotor Tricopter.

Technical Skills

Robotics and Autonomy: Soft robotic manufacturing, ROS 2, PX4 Autopilot, ArduPilot, MAVLink, Gazebo, RViz, SLAM, OFFBOARD control, AprilTag-based localization.

Programming and Control: MATLAB, Simulink, basic Python, PID control.

Embedded and Sensors: Raspberry Pi, Teensy 4.1, ESP32, IMUs, 2D LiDAR, Sensors integration.

Mechanical Design and Fabrication: SolidWorks, ANSYS, Additive manufacturing, CNC-machined.

Honors, Admissions, and Certifications

- Admitted to the Robotics, MSE Program at the University of Pennsylvania for Fall 2026 (will not enroll).
- Certified SOLIDWORKS Associate in Mechanical Design (CSWA), Dassault Systèmes, June 2026.
- Certified SOLIDWORKS Professional in Mechanical Design (CSWP), Dassault Systèmes, August 2026.
- IELTS Academic: 7.5, August 2022.